lean4ml

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Chapter 1

General Definitions and Basic Properties

Throughout this section, E is a real Hilbert space, $f : E \to \mathbb{R}$, $s \subseteq E$, and $L \in \mathbb{R}_{\geq 0}$ (Lean: `NNReal`).

Many of the properties presented here are well-known and we don't give an explicit proof. Still, these were added to our Lean project to simplify the proofs of more complex results.

1.1 L-smoothness and others

Definition 1 (L-smoothness on a set)**.** We say that f is L-smooth on s if

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \quad \forall x, y \in s.$$

Definition 2 (Global L-smoothness)**.** A function is globally L-smooth if it is L-smooth on all of E.

Lemma 3 (Global smoothness is smoothness on `Set.univ`)**.** *In Lean:*

$$\texttt{LSmoothfn f L} \iff \texttt{LSmoothOn f L Set.univ}.$$

Proof. By definition, both sides assert that ∇f is L-Lipschitz at every pair of points in E; `LSmoothOn` on `Set.univ` unfolds to the same statement. ☐

Lemma 4 (Monotonicity under set inclusion)**.** *If $T \subseteq S$ and f is L-smooth on S, then f is L-smooth on T.*

Proof. The Lipschitz inequality $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ is required for $x, y \in T$; since $T \subseteq S$, the hypothesis on S delivers it directly. ☐

Definition 5 (Hessian)**.** The Hessian is the Fréchet derivative of the gradient map:

$$\nabla^2 f(x) := D(\nabla f)(x).$$

Definition 6 (Globally μ-PL function)**.** We say that f is globally μ-PL if for all x we have

$$\|\nabla f(x)\|^2 \geq 2\mu(f(x) - \inf f).$$

1.2 Line Map and its properties

Definition 7 (Line map)**.** The canonical line map through x in direction p is $t \mapsto x + t\,p$.

Lemma 8 (Continuity of $t \mapsto (Df(x+tp))p$)**.** *If f is C^1, then $t \mapsto (Df(x+tp))p$ is continuous.*

Proof. The map $t \mapsto (x + tp, p)$ is continuous as a pair of continuous maps in t. Since $f \in C^1$, the bilinear evaluation $(y, v) \mapsto (Df(y))v$ is continuous, and the composition yields the required continuity. □

Lemma 9 (Continuity of $t \mapsto \langle \nabla f(x+tp), p\rangle$)**.** *If f is C^1, then $t \mapsto \langle \nabla f(x+tp), p\rangle$ is continuous.*

Proof. In Lean, simply rewrite $(Df(x+tp))p$ as $\langle \nabla f(x+tp), p\rangle$ using `HasGradientAt.fderiv_apply`. □

Lemma 10 (Integrability of the line integrand)**.** *The map $t \mapsto \langle\!\langle \nabla f(x + tp), p\rangle\!\rangle$ is integrable on any interval (by continuity).*

Proof. By Lemma 9 the integrand is continuous, hence bounded on any compact interval and therefore interval-integrable. □

Lemma 11 (Chain rule along a line)**.** *If f has Fréchet derivative at $x + tp$, then $\tau \mapsto f(x + \tau p)$ has derivative $(Df(x + tp))p$ at $\tau = t$.*

Proof. The map $\tau \mapsto x + \tau p$ has derivative p at $\tau = t$. Composing with f via the chain rule gives derivative $Df(x + tp) \cdot p$ at $\tau = t$. □

Lemma 12 (Convex segment stays in a convex set)**.** *If s is convex and $x, y \in s$, then $x+t(y-x) \in s$ for $t \in [0, 1]$.*

Proof. Rewrite $x + t(y - x) = (1 - t)x + ty$, which lies in s by convexity since $1 - t, t \geq 0$ and $(1 - t) + t = 1$. □

Lemma 13 (Scaling of the Hessian quadratic form)**.**

$$\langle \nabla^2 f(x)(\alpha p), \alpha p\rangle = \alpha^2 \langle \nabla^2 f(x)p, p\rangle.$$

Proof. By linearity of $\nabla^2 f(x)$ and bilinearity of the inner product, $\langle \nabla^2 f(x)(\alpha p), \alpha p\rangle = \alpha \langle \nabla^2 f(x)p, \alpha p\rangle = \alpha \cdot \alpha \, \langle \nabla^2 f(x)p, p\rangle = \alpha^2 \langle \nabla^2 f(x)p, p\rangle$. □

Lemma 14 (Hessian continuity along a line)**.** *If $f \in C^2$, then $t \mapsto \nabla^2 f(x + tp)\,p$ is continuous.*

Proof. Since $f \in C^2$, the gradient ∇f is C^1 (Riesz duality composed with Df preserves smoothness). Applying Lemma 8 to ∇f gives the result. □

1.3 Other intermediary results

Lemma 15 (Derivative operator-norm bound from Lipschitz)**.** *If g is L-Lipschitz on a set s containing a neighborhood of x and g is differentiable at x, then $\|Dg(x)\| \leq L$.*

Proof. Simply by noting that for $y \neq x$ in the neighborhood, $\frac{|f(x)-f(y)|}{\|x-y\|} \leq L$, from where the limit definition of the derivative norm gives $\|Dg(x)\| \leq L$. □

Lemma 16 (Quadratic form controlled by operator norm)**.** *For $A : E \to \mathcal{L}[\mathbb{R}]E$ and $p \in E$:*

$$\langle Ap, p\rangle \le \|A\|\,\|p\|^2.$$

Proof. By Cauchy–Schwarz, $\langle Ap, p\rangle \le \|Ap\|\,\|p\|$, and the operator-norm bound gives $\|Ap\| \le \|A\|\,\|p\|$. Multiplying yields $\langle Ap, p\rangle \le \|A\|\,\|p\|^2$. ☐

Lemma 17 (HasGradient everywhere for C^1)**.** *If $f \in C^1(E)$, then for every $z \in E$ the gradient exists.*

Proof. C^1-regularity implies Fréchet differentiability at every z. Riesz duality then identifies the derivative with an inner product against a unique element $\nabla f(z) \in E$. ☐

Lemma 18 (Norm Lipschitz bound for the gradient)**.** *If f is L-smooth on s and $x, y \in s$, then $\|\nabla f(x) - \nabla f(y)\| \le L\|x - y\|$.*

Proof. Direct from the definition of L-smoothness on s (Definition 1), unwrapping the underlying `edist` formulation to a real-valued norm inequality. ☐

Lemma 19 (Segment norm identity)**.** *For $\gamma \ge 0$, $\|(x + \gamma(y - x)) - x\| = \gamma\|y - x\|$.*

Proof. $\|(x + \gamma(y - x)) - x\| = \|\gamma(y - x)\| = |\gamma|\,\|y - x\| = \gamma\,\|y - x\|$ since $\gamma \ge 0$. ☐

Lemma 20 (Integral split-off constant term in an inner product)**.** *Lean lemma used to rewrite $\int\langle g(t) - c, w\rangle = \int\langle g(t), w\rangle - \int\langle c, w\rangle$.*

Proof. By linearity of the inner product in the first argument, $\langle g(t) - c, w\rangle = \langle g(t), w\rangle - \langle c, w\rangle$. Both terms are interval-integrable (the first by assumption, the second since it is constant), so linearity of the integral gives the identity. ☐

Lemma 21 (Monotonicity of interval integrals (continuous case))**.** *If $\varphi \le \psi$ on $[a, b]$ with $a \le b$ and both are continuous, then $\int_a^b \varphi \le \int_a^b \psi$.*

Proof. Continuity on $[a, b]$ ensures both functions are interval-integrable. Standard monotonicity of the Lebesgue integral over the ordered interval $[a, b]$ then gives the inequality. ☐

Lemma 22 (Inner-product linearity rewrite)**.** $\langle u, w\rangle - \langle v, w\rangle = \langle u - v, w\rangle$.

Proof. Linearity of the inner product in its first argument. ☐

Chapter 2

Taylor's Theorem

Theorem 23 (First-order Taylor formula along a line (vector-valued))**.** *If $f : E \to F$ is C^1, then*

$$f(x+p) - f(x) = \int_0^1 (Df(x+tp))p\, dt.$$

Proof. Let $g(t) := f(x + t\,p) \quad (t \in \mathbb{R})$. We apply the fundamental theorem of calculus to g on $[0, 1]$.

For each t, since $f \in C^1$, f is Fréchet differentiable at $x + t\,p$, with derivative $Df(x + t\,p)$. By the chain rule along a line (Lemma 11), $g'(t) = (Df(x + t\,p))p$. Hence g has the required derivative at every point of $\mathrm{uIcc}(0, 1)$.

Next, the integrand $t \mapsto (Df(x+t\,p))p$ is interval-integrable on $[0, 1]$ by continuity (Lemma 8).

Therefore, by `intervalIntegral.integral_eq_sub_of_hasDerivAt`,

$$\int_0^1 (Df(x + t\,p))p\, dt = g(1) - g(0).$$

Expanding $g(1) = f(x + p)$ and $g(0) = f(x)$ gives

$$f(x+p) - f(x) = \int_0^1 (Df(x + t\,p))p\, dt.$$

□

Theorem 24 (First-order Taylor formula along a line)**.** *If $f \in C^1$, then*

$$f(x+p) - f(x) = \int_0^1 \langle \nabla f(x+tp), p \rangle\, dt.$$

Proof. Direct from Theorem 23. □

Theorem 25 (Second-order Taylor formula for the gradient)**.** *If $f \in C^2$, then*

$$\nabla f(x+p) - \nabla f(x) = \int_0^1 \nabla^2 f(x+tp)\, p\, dt.$$

Proof. Direct from Theorem 23 applied to $\nabla f \in C^1$. □

Lemma 26 (Pointwise Lipschitz-gradient bound)**.** *For $f \in C^1$ that is L-smooth, for $\gamma \geq 0$ and the segment point in s,*

$$\langle \nabla f(x + \gamma(y - x)) - \nabla f(x), y - x\rangle \leq L\gamma\|y - x\|^2.$$

Proof. First, derive a norm bound from L-smoothness. Since $x \in s$ and $x + \gamma(y - x) \in s$, Lemma 18 gives

$$\|\nabla f(x + \gamma(y - x)) - \nabla f(x)\| \leq L\,\|(x + \gamma(y - x)) - x\|.$$

Using Lemma 19 (with $\gamma \geq 0$), $\|(x+\gamma(y-x))-x\| = \gamma\|y-x\|$, hence $\|\nabla f(x+\gamma(y-x))-\nabla f(x)\| \leq L\gamma\|y - x\|$.

Second, apply Cauchy-Schwarz:

$$\langle \nabla f(x + \gamma(y - x)) - \nabla f(x), y - x\rangle \leq \|\nabla f(x + \gamma(y - x)) - \nabla f(x)\|\,\|y - x\|.$$

Plug the previous bound into the right-hand side to get

$$\langle \nabla f(x + \gamma(y - x)) - \nabla f(x), y - x\rangle \leq (L\gamma\|y - x\|)\|y - x\| = L\gamma\|y - x\|^2,$$

which is exactly the claim. □

Lemma 27 (Non-parametric corollary ($\gamma = 1$))**.** *In particular, if $x, y \in s$ then $\langle \nabla f(y) - \nabla f(x), y - x\rangle \leq L\|y - x\|^2$.*

Proof. Set $\gamma = 1$ in Lemma 26 and note that $x + 1 \cdot (y - x) = y$. □

Theorem 28 (Quadratic upper bound from smoothness on a convex set)**.** *If $f \in C^1$, f is L-smooth on convex s, and $x, y \in s$, then*

$$f(y) \leq f(x) + \langle \nabla f(x), y - x\rangle + \frac{L}{2}\|y - x\|^2.$$

Proof. Let $p := y - x$.

1. Apply Theorem 24 to obtain $f(y) - f(x) = \int_0^1 \langle \nabla f(x + tp), p\rangle\, dt$.
2. Rewrite $f(y) - f(x) - \langle \nabla f(x), p\rangle$ as an integral of $\langle \nabla f(x + tp) - \nabla f(x), p\rangle$ using Lemma 20.
3. Use convexity via Lemma 12 to ensure $x + tp \in s$ for $t \in [0, 1]$.
4. Bound the integrand pointwise using Lemma 26 with $\gamma = t$, then integrate the bound with Lemma 21.
5. Evaluate $\int_0^1 Lt\, dt = L/2$ and conclude.

□

Theorem 29 (Descent Lemma for Gradient Descent)**.** *If $f \in C^1$, f is L-smooth on a convex set s, $x \in s$, $\alpha \geq 0$, and $x - \alpha\nabla f(x) \in s$, then*

$$f(x - \alpha\nabla f(x)) \leq f(x) - \alpha\left(1 - \frac{L\alpha}{2}\right)\|\nabla f(x)\|^2.$$

Proof. Let $y := x - \alpha \nabla f(x)$ and $p := y - x = -\alpha \nabla f(x)$.

Step 1: Compute the inner product.

$$\langle \nabla f(x), p \rangle = \langle \nabla f(x), -\alpha \nabla f(x) \rangle = -\alpha \|\nabla f(x)\|^2.$$

Step 2: Compute the norm squared of p using $\alpha \geq 0$ to simplify $|\alpha| = \alpha$.

$$\|p\|^2 = \| - \alpha \nabla f(x)\|^2 = (\|\alpha \nabla f(x)\|)^2 = \alpha^2 \|\nabla f(x)\|^2.$$

Step 3: Apply Theorem 28 with the points x and $y = x + p$:

$$f(y) \leq f(x) + \langle \nabla f(x), p \rangle + \frac{L}{2}\|p\|^2.$$

Step 4: Substitute the expressions from Steps 1 and 2:

$$\begin{aligned} f(y) &\leq f(x) + (-\alpha \|\nabla f(x)\|^2) + \frac{L}{2}(\alpha^2 \|\nabla f(x)\|^2) \\ &= f(x) - \alpha \|\nabla f(x)\|^2 + \frac{L\alpha^2}{2}\|\nabla f(x)\|^2 \\ &= f(x) - \alpha \left(1 - \frac{L\alpha}{2}\right) \|\nabla f(x)\|^2. \end{aligned}$$

This completes the proof. $\square$

Chapter 3

Hessian characterization of L-Smooth functions

Theorem 30 (Hessian operator-norm bound $\Rightarrow$ global smoothness)**.** *If ∇f is differentiable everywhere with derivative $\nabla^2 f(x)$, and $\|\nabla^2 f(x)\| \leq L$ for all x, then f is globally L-smooth.*

Proof. We need to show ∇f is L-Lipschitz globally.

Since ∇f is Fréchet differentiable at every point x with derivative $\nabla^2 f(x)$ (by hypothesis), and we have the operator-norm bound $\|\nabla^2 f(x)\| \leq L$ for all x, Mathlib's `lipschitzWith_of_nnnorm_fderiv_le` directly gives that ∇f is L-Lipschitz on all of E. By definition, this means f is globally L-smooth. ☐

Theorem 31 (Global smoothness $\Rightarrow$ Hessian quadratic-form upper bound)**.** *If f is globally L-smooth, then $\langle \nabla^2 f(x)p, p\rangle \leq L\|p\|^2$ for all x, p.*

Proof. Fix x and p. By global L-smoothness, ∇f is L-Lipschitz on all of E.

Apply Lemma 15 to the gradient map: since ∇f is L-Lipschitz on a neighborhood of x (in fact, globally) and is Fréchet differentiable at x with derivative $\nabla^2 f(x)$, we obtain

$$\|\nabla^2 f(x)\| \leq L.$$

By Lemma 16 (operator norm controls the quadratic form),

$$\langle \nabla^2 f(x)p, p\rangle \leq \|\nabla^2 f(x)\| \cdot \|p\|^2 \leq L\|p\|^2,$$

as required. ☐

Chapter 4

NecessaryConditions

Theorem 32 (Local minimum $\Rightarrow$ gradient vanishes)**.** *If $f \in C^1$ and x is a local minimizer (Lean:* `IsLocalMin f x`*), then $\nabla f(x) = 0$.*

Proof. By Lemma 17, the gradient $\nabla f(x)$ exists at the local minimizer x.

Since x minimizes f locally, the Fréchet derivative of f at x must vanish when applied to any direction. The Fréchet derivative of f at x is the inner product with $\nabla f(x)$, so we have

$$\langle \nabla f(x), v \rangle = 0 \quad \text{for all } v \in E.$$

Setting $v = \nabla f(x)$ gives $\|\nabla f(x)\|^2 = 0$, hence $\nabla f(x) = 0$. $\square$

Lemma 33 (Gradient as integral of Hessian along a line)**.** *If $f \in C^2$ and $\nabla f(x) = 0$, then $\nabla f(x + tp) = \int_0^t \nabla^2 f(x + sp)\, p\, ds$.*

Proof. Apply Theorem 25 with direction tp to get $\nabla f(x+tp) - \nabla f(x) = \int_0^1 \nabla^2 f(x+u\,tp)(tp)\, du$. Using $\nabla f(x) = 0$ and the linear substitution $s = ut$, this becomes $\int_0^t \nabla^2 f(x + sp)\, p\, ds$. $\square$

Lemma 34 (Exact second-order Taylor expansion (double integral))**.** *If $f \in C^2$ and $\nabla f(x) = 0$, then*

$$f(x + p) - f(x) = \int_0^1 \int_0^t \langle \nabla^2 f(x + sp)p, p \rangle \, ds\, dt.$$

Proof. By Theorem 24, $f(x + p) - f(x) = \int_0^1 \langle \nabla f(x + tp), p \rangle\, dt$. Using Lemma 33, $\nabla f(x + tp) = \int_0^t \nabla^2 f(x + sp)p\, ds$. Substituting and exchanging the inner product with the (continuous-linear) integral yields the double-integral formula. $\square$

Theorem 35 (Local minimum $\Rightarrow$ Hessian PSD)**.** *If $f \in C^2$ and x is a local minimizer, then $\langle \nabla^2 f(x)p, p \rangle \geq 0$ for all p.*

Proof. We proceed by contradiction. First, by Theorem 32, we may assume $\nabla f(x) = 0$.

Suppose $\langle \nabla^2 f(x)p, p \rangle < 0$ for some $p \neq 0$. Define $H(u) := \langle \nabla^2 f(x + up)p, p \rangle$. Then $H(0) = \langle \nabla^2 f(x)p, p \rangle < 0$. By continuity of the Hessian (Lemma 14), there exists $\delta > 0$ such that $H(u) < H(0)/2 < 0$ for all $u \in (-\delta, \delta)$.

But x is a local minimizer, so $f(x + \alpha p) \geq f(x)$ for small $\alpha > 0$. By the exact second-order Taylor formula (Lemma 34),

$$f(x + \alpha p) - f(x) = \int_0^1 \int_0^t \langle \nabla^2 f(x + s\alpha p)p, p\rangle \, ds \, dt = \alpha^2 \int_0^1 \int_0^t \langle \nabla^2 f(x + s\alpha p)p, p\rangle \, ds \, dt.$$

For small $\alpha > 0$, all points $x + s\alpha p$ lie in the ball where $H < 0$, so both integrals are negative, making $f(x + \alpha p) - f(x) < 0$, contradicting local minimality. $\square$

Chapter 5

Sufficient Conditions

Definition 36 (Strict local minimizer)**.** Lean definition: $\exists U \in \mathcal{N}(x),\ \forall y \in U,\ y \neq x \to f(x) < f(y)$.

Lemma 37 (Taylor lower bound from uniform Hessian lower bound)**.** *Under a uniform bound $\varepsilon\|p\|^2 \leq \langle \nabla^2 f(y)p, p\rangle$ on a ball around x and assuming $\nabla f(x) = 0$, Lean proves $(\varepsilon/2)\|p\|^2 \leq f(x+p) - f(x)$ for $\|p\| < \rho$ by integrating the pointwise bound inside the double-integral remainder formula.*

Proof. Start from the exact Taylor formula (Lemma 34):

$$f(x+p) - f(x) = \int_0^1 \int_0^t \langle \nabla^2 f(x+sp)p, p\rangle \, ds \, dt.$$

By convexity (or simply by norm estimates), the point $x + sp$ stays within the ball $B(x, \rho)$ for all $s \in [0, 1]$ when $\|p\| < \rho$. Within this ball, the Hessian quadratic form is bounded below: $\langle \nabla^2 f(y)p, p\rangle \geq \varepsilon\|p\|^2$.

Therefore,

$$f(x+p) - f(x) = \int_0^1 \int_0^t \langle \nabla^2 f(x+sp)p, p\rangle \, ds \, dt \geq \int_0^1 \int_0^t \varepsilon\|p\|^2 \, ds \, dt = \varepsilon\|p\|^2 \int_0^1 t \, dt = \frac{\varepsilon}{2}\|p\|^2,$$

as claimed. □

Lemma 38 (Coercivity from positive definiteness (finite-dimensional))**.** *In finite dimensions, a positive definite continuous linear map H satisfies $\exists \varepsilon > 0,\ \forall p,\ \varepsilon\|p\|^2 \leq \langle Hp, p\rangle$. Lean uses compactness of the sphere and `IsCompact.exists_isLeast`.*

Proof. In finite dimensions the unit sphere $S = \{u : \|u\| = 1\}$ is compact. The continuous function $u \mapsto \langle Hu, u\rangle$ attains its minimum ε on S, and $\varepsilon > 0$ by positive definiteness. For an arbitrary $p \neq 0$, writing $p = \|p\| \cdot (p/\|p\|)$ and using bilinearity gives $\langle Hp, p\rangle = \|p\|^2 \langle H(p/\|p\|), p/\|p\|\rangle \geq \varepsilon\|p\|^2$. The inequality holds trivially at $p = 0$. □

Lemma 39 (Continuity of Hessian map)**.** *If $f \in C^2$, then $y \mapsto \nabla^2 f(y)$ is continuous (Lean: `continuous_fderiv` applied to ∇f).*

Proof. Since $f \in C^2$, the gradient ∇f is C^1, so its Fréchet derivative $y \mapsto D(\nabla f)(y) = \nabla^2 f(y)$ is continuous. □

Lemma 40 (Positive definite Hessian $\Rightarrow$ uniform lower bound on a ball)**.** *In finite dimensions, if $\nabla^2 f(x)$ is positive definite, then there exist $\varepsilon > 0$ and $\rho > 0$ such that $\varepsilon\|p\|^2 \leq \langle \nabla^2 f(y)p, p\rangle$ for all $y \in B(x, \rho)$ and all p.*

Proof. Since $\nabla^2 f(x)$ is positive definite, Lemma 38 provides $\varepsilon_0 > 0$ such that

$$\varepsilon_0\|p\|^2 \leq \langle \nabla^2 f(x)p, p\rangle \quad \text{for all } p.$$

By Lemma 39, the map $y \mapsto \nabla^2 f(y)$ is continuous. Therefore, we can find $\rho > 0$ such that

$$\|\nabla^2 f(y) - \nabla^2 f(x)\| < \frac{\varepsilon_0}{2} \quad \text{for all } y \in B(x, \rho).$$

For $y \in B(x, \rho)$ and any p, the operator norm inequality gives

$$\left|\langle(\nabla^2 f(y) - \nabla^2 f(x))p, p\rangle\right| \leq \|\nabla^2 f(y) - \nabla^2 f(x)\|\|p\|^2 < \frac{\varepsilon_0}{2}\|p\|^2.$$

Therefore,

$$\langle \nabla^2 f(y)p, p\rangle = \langle \nabla^2 f(x)p, p\rangle + \langle(\nabla^2 f(y) - \nabla^2 f(x))p, p\rangle > \varepsilon_0\|p\|^2 - \frac{\varepsilon_0}{2}\|p\|^2 = \frac{\varepsilon_0}{2}\|p\|^2,$$

which completes the proof. □

Theorem 41 (Strict local minimizer from uniform Hessian lower bound)**.** *If $f \in C^2$, $\nabla f(x) = 0$, and the Hessian has a uniform positive lower bound on a ball around x, then x is a strict local minimizer.*

Proof. Let $U = B(x, \rho)$. For any $y \in U$ with $y \neq x$, set $p = y - x$. Then $\|p\| < \rho$ and $f(y) - f(x) = f(x + p) - f(x)$.

By Lemma 37 (applied to the uniform lower bound on $\nabla^2 f$ over the ball around x), we have

$$f(x + p) - f(x) \geq \frac{\varepsilon}{2}\|p\|^2.$$

Since $p \neq 0$, we have $\|p\|^2 > 0$, so $f(y) > f(x)$. This holds for all $y \in U$ with $y \neq x$, showing that x is a strict local minimizer. □

Theorem 42 (Second-order sufficient condition (finite-dimensional))**.** *If $f \in C^2$, $\nabla f(x) = 0$, and $\nabla^2 f(x)$ is positive definite, then x is a strict local minimizer.*

Proof. By hypothesis, $f \in C^2$ and $\nabla f(x) = 0$. The Hessian $\nabla^2 f(x)$ is positive definite.

Lemma 40 guarantees that there exist $\varepsilon > 0$ and $\rho > 0$ such that

$$\varepsilon\|p\|^2 \leq \langle \nabla^2 f(y)p, p\rangle \quad \text{for all } y \in B(x, \rho) \text{ and all } p.$$

With this uniform lower bound in place, Theorem 41 applies immediately to conclude that x is a strict local minimizer. □

Chapter 6

GradientFlow

Throughout this section, E is a real Hilbert space, $f : E \to \mathbb{R}$, and ∇f denotes the (Fréchet) gradient in the Hilbert space structure.

Definition 43 (Gradient flow on a bounded interval)**.** A *bounded-time* gradient flow is a certificate that the ODE

$$\dot{x}(t) = -\nabla f(x(t)), \qquad x(0) = x_0$$

has a solution on $[0, T]$. In Lean, a `GradientFlow f x  T` is a structure with fields:

- `trajectory :` $\mathbb{R} \to E$,
- `init : trajectory 0 = x` ,
- `hasDerivAt :   t   Icc 0 T, HasDerivAt trajectory (-(gradient f (trajectory t))) t`.

Lemma 44 (Initial value)**.** *For* φ : ***`GradientFlow f x  T`***, *we have* $\varphi(0) = x_0$.

Lemma 45 (Differentiability of the trajectory on $[0, T]$)**.** *For* $t \in [0, T]$, *the map* $t \mapsto \varphi(t)$ *is differentiable at* t. *(Lean:* ***`DifferentiableAt   .trajectory t`***.*)*

Lemma 46 (Derivative equals negative gradient)**.** *For* $t \in [0, T]$,

$$\varphi'(t) = -\nabla f(\varphi(t)).$$

(Lean: ***`deriv  .trajectory t = -(gradient f ( .trajectory t))`***.*)*

Lemma 47 (Energy derivative along the flow)**.** *If* $f \in C^1$ *and* φ *is a gradient flow, then for* $t \in [0, T]$

$$\frac{d}{dt} f(\varphi(t)) = -\|\nabla f(\varphi(t))\|^2.$$

Proof. By the chain rule, the derivative of the composition $f \circ \varphi$ at time t is

$$\frac{d}{dt} f(\varphi(t)) = Df(\varphi(t)) \cdot \varphi'(t) = \langle \nabla f(\varphi(t)), \varphi'(t) \rangle.$$

Substituting the ODE $\varphi'(t) = -\nabla f(\varphi(t))$ gives

$$\frac{d}{dt} f(\varphi(t)) = \langle \nabla f(\varphi(t)), -\nabla f(\varphi(t)) \rangle = -\|\nabla f(\varphi(t))\|^2,$$

as claimed. □

Lemma 48 (Energy is non-increasing). *If $0 \le r \le s \le T$, then*

$$f(\varphi(s)) \le f(\varphi(r)).$$

Proof. Suppose for contradiction that $f(\varphi(s)) > f(\varphi(r))$. By the mean value theorem, there exists $t \in (r, s)$ such that

$$\varphi'(t) = \frac{f(\varphi(s)) - f(\varphi(r))}{s - r} > 0.$$

But by Lemma 47, we have

$$\varphi'(t) = -\|\nabla f(\varphi(t))\|^2 \le 0,$$

a contradiction. Thus $f(\varphi(s)) \le f(\varphi(r))$. □

Definition 49 (Piecewise trajectory). The piecewise trajectory that follows φ_1 up to time T and then φ_2 (shifted) is

$$\Phi(t) := \begin{cases} \varphi_1(t), & t \le T, \\ \varphi_2(t - T), & t > T. \end{cases}$$

(Lean: `if t   T then   t else   (t - T)`.)

Definition 50 (Extend (chain) two gradient flows). Given a flow φ on $[0, T]$ and a flow ψ on $[0, \delta]$ starting at $\varphi(T)$, the piecewise trajectory (Definition 49) forms a gradient flow on $[0, T + \delta]$. The derivative condition holds throughout: it matches the flow equations on $(0, T)$ and on $(T, T + \delta)$, and at the junction point $t = T$, the derivatives agree by the initial condition of ψ.

Definition 51 (Restrict a flow to a smaller time interval). A flow on $[0, S]$ restricts to one on $[0, T]$ for $T \le S$ by reusing the same trajectory and weakening the domain proof.

Lemma 52 (Gradient map is continuous). *If $f \in C^1$, then ∇f is continuous by the definition of the gradient in terms of the Fréchet derivative.*

Lemma 53 (Norm bound from a Lipschitz gradient). *If ∇f is L-Lipschitz, then by the reverse triangle inequality,*

$$\|\nabla f(x)\| \le \|\nabla f(y)\| + \|\nabla f(x) - \nabla f(y)\| \le \|\nabla f(y)\| + L\|x - y\|.$$

Theorem 54 (Local gradient flow existence). *If $f \in C^1$ and ∇f is globally L-Lipschitz, then for any x_0 there exists $\delta > 0$ and a gradient flow on $[0, \delta]$ starting at x_0. This is a direct application of the Picard–Lindelöf existence theorem for ODEs to the vector field $x \mapsto -\nabla f(x)$.*

Definition 55 (Concrete local flow constructor). A concrete local flow can be constructed explicitly on the time step

$$\delta := \frac{1}{4L + 4}.$$

This formula comes from applying the Picard–Lindelöf theorem with careful tracking of the constants in the problem.

Theorem 56 (Uniform local existence). *There exists $\delta > 0$ (in fact $\delta = 1/(4L + 4)$) such that for* every *x_0 there is a gradient flow on $[0, \delta]$ starting from x_0.*

Lemma 57 (Iterated extension). *From the local existence result (Theorem 54) with the uniform time step $\delta = 1/(4L+4)$, we can construct flows on $[0, n\delta]$ for any $n \in \mathbb{N}$ by repeatedly extending the flow at each time interval $[(k - 1)\delta, k\delta]$.*

Theorem 58 (Gradient flow existence on $[0, T]$)**.** *If $f \in C^1$, ∇f is globally L-Lipschitz, and $T > 0$, then for any x_0 there exists a gradient flow on $[0, T]$ starting at x_0.*

Proof. We choose $\delta = 1/(4L + 4)$ from Theorem 56. Let $n = \lceil T/\delta \rceil + 1$, so that $T \leq n\delta$. By Lemma 57, we obtain a gradient flow on $[0, n\delta]$. Finally, by Definition 51, we restrict to $[0, T]$ to get the desired flow. □

Definition 59 (Global gradient flow)**.** A global gradient flow for initial condition x_0 is a trajectory $\varphi : \mathbb{R} \to \mathcal{H}$ with $\varphi(0) = x_0$ and satisfying the ODE $\varphi'(t) = -\nabla f(\varphi(t))$ for all $t \geq 0$.

Definition 60 (Restriction of a global flow)**.** A global flow restricts to a bounded flow on $[0, T]$ by reusing the same trajectory.

Lemma 61 (Uniqueness of bounded flows)**.** *If ∇f is L-Lipschitz and $T \geq 0$, then two gradient flows with the same initial condition must coincide on $[0, T]$. This follows from the standard ODE uniqueness theorem applied to the Lipschitz vector field $x \mapsto -\nabla f(x)$.*

Lemma 62 (Consistency at overlaps)**.** *For a family of flows $\varphi(n)$ on $[0, n + 1]$, any two indexes agree at any time t in both domains. This follows from uniqueness: the two flows satisfy the same ODE with the same initial data, so they must coincide on the overlap.*

Lemma 63 (Local agreement of the glued global trajectory)**.** *Define the global trajectory by $x(t) := (\varphi(\lceil t \rceil))$.trajectory$(t)$. For each $t \geq 0$, near t this trajectory agrees with the bounded flow $(\varphi(\lceil t \rceil))$.trajectory, so the derivative at t transfers from the bounded flow.*

Theorem 64 (Global gradient flow existence for all $t \geq 0$)**.** *If $f \in C^1$ and ∇f is globally L-Lipschitz, then for any x_0 there exists a global gradient flow defined for all $t \geq 0$.*

Proof. For each $n \in \mathbb{N}$, apply Theorem 58 to obtain a bounded gradient flow $\varphi(n)$ on $[0, n+1]$. Define the global trajectory by $x(t) := (\varphi(\lceil t \rceil))$.trajectory$(t)$ for all $t \geq 0$. By Lemma 63, for each $t \geq 0$, the trajectory x agrees locally with $\varphi(\lceil t \rceil)$ near t. Therefore, x satisfies the ODE for all $t \geq 0$ by transfer of derivatives from the local agreement. □

Chapter 7

Convexity

Throughout this section, E is a real normed space, $f : E \to \mathbb{R}$, $s \subseteq E$ is a convex set, and $x, y \in E$.

Definition 65 (Convex function on a set)**.** We say that f is *convex on s* if s is a convex set and for all $x, y \in s$ and $a, b \geq 0$ with $a + b = 1$:

$$f(ax + by) \;\leq\; a\, f(x) + b\, f(y).$$

Theorem 66 (Local minimizer on convex set is global)**.** *If f is convex on s and $x \in s$ is a local minimizer of f restricted to s, then x is a global minimizer of f on s.*

Theorem 67 (Set of minimizers is convex)**.** *If f is convex on s, then the set of minimizers*

$$\mathcal{M} := \{x \in s \mid \forall y \in s,\; f(x) \leq f(y)\}$$

is convex.

Proof. Let $x, y \in \mathcal{M}$ and $a, b \geq 0$ with $a + b = 1$. Set $z := ax + by$. Since s is convex, $z \in s$. For any $w \in s$, convexity of f gives

$$f(z) \leq a\, f(x) + b\, f(y).$$

Since x and y are both minimizers, $f(x) \leq f(w)$ and $f(y) \leq f(w)$, so

$$f(z) \leq a\, f(x) + b\, f(y) \leq a\, f(w) + b\, f(w) = (a + b)\, f(w) = f(w).$$

Hence $z \in \mathcal{M}$. If $\mathcal{M}$ is empty, it is trivially convex. $\square$

Lemma 68 (First-order characterization of convexity)**.** *If f is convex on s and Fréchet differentiable at $x \in s$, then for all $y \in s$:*

$$f(x) + Df(x)(y - x) \leq f(y).$$

Proof. **Step 1: Convexity upper bound along the segment.** For $\alpha \in [0, 1]$, the convexity of f gives

$$f(x + \alpha(y - x)) = f((1 - \alpha)x + \alpha y) \leq (1 - \alpha)f(x) + \alpha f(y).$$

Step 2: Rearranging into a difference quotient. For $\alpha \in (0, 1]$, subtracting $f(x)$ and dividing by $\alpha > 0$ yields

$$\frac{f(x + \alpha(y - x)) - f(x)}{\alpha} \leq f(y) - f(x).$$

Step 3: Taking the limit. By the chain rule, the map $\alpha \mapsto f(x + \alpha(y - x))$ has derivative $Df(x)(y - x)$ at $\alpha = 0$. Therefore the difference quotients on the left converge to $Df(x)(y - x)$ as $\alpha \to 0^+$. Passing the limit through the inequality gives

$$Df(x)(y - x) \leq f(y) - f(x),$$

which rearranges to $f(x) + Df(x)(y - x) \leq f(y)$. □

Theorem 69 (Zero derivative implies global minimizer)**.** *If f is convex on s, $x \in s$, and $Df(x) = 0$, then x is a global minimizer of f on s.*

Proof. Apply Lemma 68 with the given x and any $y \in s$:

$$f(x) + Df(x)(y - x) \leq f(y).$$

Since $Df(x) = 0$, the second term vanishes, giving $f(x) \leq f(y)$ for all $y \in s$. □

Chapter 8

Polyak–Łojasiewicz condition and convergence

Lemma 70 (Energy gap derivative and PL differential inequality)**.** *Let $g(t) := f(\varphi(t)) - \inf f$, where φ is a global gradient flow. If $f \in C^1$ and f satisfies the global μ-PL condition, then for every $t \geq 0$,*

$$g'(t) = -\|\nabla f(\varphi(t))\|^2 \quad \textit{and} \quad g'(t) \leq -2\mu\, g(t).$$

Proof. Differentiate $g(t) = f(\varphi(t)) - \inf f$. The constant $\inf f$ contributes no derivative, and the chain rule along the flow gives

$$g'(t) = \frac{d}{dt} f(\varphi(t)) = -\|\nabla f(\varphi(t))\|^2.$$

The global PL inequality gives

$$\|\nabla f(\varphi(t))\|^2 \geq 2\mu(f(\varphi(t)) - \inf f) = 2\mu\, g(t),$$

so multiplying by -1 yields $g'(t) \leq -2\mu\, g(t)$. □

Lemma 71 (Exponential decay of the energy gap)**.** *Under the same hypotheses, the energy gap satisfies*

$$f(\varphi(t)) - \inf f \leq (f(x_0) - \inf f)\, e^{-2\mu t}$$

for all $t \geq 0$.

Proof. Consider the auxiliary function

$$h(t) := (f(\varphi(t)) - \inf f)e^{2\mu t}.$$

Using the previous lemma and the product rule, one finds that $h'(t) \leq 0$ for $t \geq 0$, so h is non-increasing. Therefore $h(t) \leq h(0) = f(x_0) - \inf f$, and multiplying by $e^{-2\mu t}$ gives the stated estimate. □

Theorem 72 (Convergence of the gradient flow under the PL condition)**.** *Assume in addition that f is bounded below. Then along any global gradient flow,*

$$f(\varphi(t)) \to \inf f \quad \textit{as } t \to \infty.$$

Proof. The global PL estimate gives the upper bound from Lemma 71. Since f is bounded below, we also have $f(\varphi(t)) - \inf f \geq 0$ for all $t \geq 0$. The right-hand side is an exponentially decaying term, hence tends to 0 as $t \to \infty$. By squeezing, the energy gap converges to 0, which is equivalent to $f(\varphi(t)) \to \inf f$. $\square$

Chapter 9

Strong Convexity

Throughout this section, E is a real Hilbert space, $f : E \to \mathbb{R}$, $s \subseteq E$ is a convex set, $m > 0$, and $x, y \in E$.

Definition 73 (Strongly convex function on a set)**.** We say that f is *strongly convex on s with modulus $m > 0$* if s is a convex set and for all $x, y \in s$ and $a, b \geq 0$ with $a + b = 1$:

$$f(ax + by) \;\leq\; a\, f(x) + b\, f(y) - \tfrac{m}{2}\, ab\, \|x - y\|^2.$$

In particular, taking $a = 1 - \alpha$, $b = \alpha$ recovers the modulated convexity inequality

$$f((1-\alpha)x + \alpha y) \;\leq\; (1-\alpha) f(x) + \alpha f(y) - \tfrac{m}{2}\alpha(1-\alpha)\|x - y\|^2.$$

Lemma 74 (Strong convexity reduces to convexity)**.** *f is strongly convex on s with modulus m if and only if $x \mapsto f(x) - \dfrac{m}{2}\|x\|^2$ is convex on s:*

$$\text{StrongConvexOn } s\ m\ f \iff \text{ConvexOn } s\ \Big(x \mapsto f(x) - \tfrac{m}{2}\|x\|^2\Big).$$

Lemma 75 (First-order lower bound for strongly convex functions)**.** *If f is strongly convex on s with modulus $m > 0$ and Fréchet differentiable at $x \in s$, then for all $y \in s$:*

$$f(y) \geq f(x) + Df(x)(y - x) + \frac{m}{2}\|y - x\|^2.$$

Proof. Define $g : E \to \mathbb{R}$ by $g(z) = f(z) - \frac{m}{2}\|z\|^2$. By Lemma 74, g is convex on s. Since f is Fréchet differentiable at x and $z \mapsto \frac{m}{2}\|z\|^2$ is differentiable everywhere, g is Fréchet differentiable at x with

$$Dg(x) = Df(x) - D\big(\tfrac{m}{2}\|\cdot\|^2\big)\,(x).$$

A direct computation gives $D\big(\frac{m}{2}\|\cdot\|^2\big)\,(x)(v) = m\langle x, v\rangle$ for all v. Apply Lemma 68 to the convex function g:

$$g(x) + Dg(x)(y - x) \leq g(y).$$

Substituting $g(z) = f(z) - \frac{m}{2}\|z\|^2$ and $Dg(x)(v) = Df(x)(v) - m\langle x, v\rangle$:

$$f(x) - \tfrac{m}{2}\|x\|^2 + Df(x)(y - x) - m\langle x,\, y - x\rangle \;\leq\; f(y) - \tfrac{m}{2}\|y\|^2.$$

Rearranging, it remains to show that

$$-\tfrac{m}{2}\|x\|^2 - m\langle x,\, y - x\rangle + \tfrac{m}{2}\|y\|^2 \;=\; \tfrac{m}{2}\|y - x\|^2.$$

Indeed, expanding $\langle x, y-x\rangle = \langle x, y\rangle - \|x\|^2$:

$$\begin{aligned}-\tfrac{m}{2}\|x\|^2 - m\langle x, y\rangle + m\|x\|^2 + \tfrac{m}{2}\|y\|^2 &= \tfrac{m}{2}\|x\|^2 - m\langle x, y\rangle + \tfrac{m}{2}\|y\|^2 \\ &= \tfrac{m}{2}(\|x\|^2 - 2\langle x, y\rangle + \|y\|^2) = \tfrac{m}{2}\|y-x\|^2.\end{aligned}$$

Therefore $f(x) + Df(x)(y-x) + \frac{m}{2}\|y-x\|^2 \le f(y)$. ☐

Theorem 76 (Unique global minimizer under strong convexity)**.** *Suppose that f is strongly convex on s with modulus $m > 0$ and Fréchet differentiable at $x^* \in s$. If $Df(x^*) = 0$, then x^* is a global minimizer of f on s, and any other global minimizer $y \in s$ satisfies $y = x^*$.*

Proof. Apply Lemma 75 with $x = x^*$ and any y in the domain:

$$f(y) \ge f(x^*) + Df(x^*)(y-x^*) + \frac{m}{2}\|y-x^*\|^2.$$

Since x^* is a stationary point, $Df(x^*) = 0$, so:

$$f(y) \ge f(x^*) + \frac{m}{2}\|y-x^*\|^2.$$

By the definition of strong convexity, the modulus $m > 0$. Furthermore, for any $y \neq x^*$, the squared norm $\|y-x^*\|^2 > 0$. Therefore, for all $y \neq x^*$, the term $\frac{m}{2}\|y-x^*\|^2$ is strictly positive, which implies:

$$f(y) > f(x^*)$$

This demonstrates that x^* is the strictly unique global minimizer of f. ☐

Lemma 77 (C^1 regularity of the gradient)**.** *If $f \in C^2$, then $y \mapsto \nabla f(y)$ is in C^1.*

Lemma 78 (Hessian as Fréchet derivative of the gradient)**.** *If $f \in C^2$, then at every $x \in E$, the map $y \mapsto \nabla f(y)$ has Fréchet derivative $\nabla^2 f(x)$.*

Lemma 79 (Strong convexity $\Rightarrow$ Hessian lower bound)**.** *If $f \in C^2$ is strongly convex on E with modulus m, then*

$$m\|u\|^2 \;\le\; \langle \nabla^2 f(x)\, u,\, u\rangle \quad \textit{for all } x, u \in E.$$

Proof. Fix x, u and $t > 0$. The first-order strong convexity bound (Lemma 75) at the pair $(x, x+tu)$ and at $(x+tu, x)$ together with summation yields

$$\frac{(Df(x+tu) - Df(x))(u)}{t} \;\ge\; m\|u\|^2.$$

As $t \to 0^+$, the left-hand side converges to $\langle \nabla^2 f(x)u, u\rangle$ (using Lemma 78), giving the claim. ☐

Lemma 80 (Hessian lower bound $\Rightarrow$ strong convexity)**.** *If $f \in C^2$ and $m\|u\|^2 \le \langle \nabla^2 f(x)u, u\rangle$ for all $x, u \in E$, then f is strongly convex on E with modulus m.*

Proof. By Lemma 74, it suffices to show that $g(y) := f(y) - \frac{m}{2}\|y\|^2$ is convex on E. For any line restriction $\varphi(t) := g(x+td)$, a direct computation gives $\varphi''(t) = \langle \nabla^2 f(x+td)\, d, d\rangle - m\|d\|^2 \ge 0$, so φ is convex. Since this holds for every line, g is convex. ☐

Lemma 81 (Hessian characterization of strong convexity)**.** *Suppose $f \in C^2$. Then f is strongly convex on E with modulus m if and only if $m\|u\|^2 \le \langle \nabla^2 f(x)u, u\rangle$ for all $x, u \in E$, i.e. $\nabla^2 f(x) \succeq mI$.*

Proof. Direct from Lemmas 79 and 80. □

Lemma 82 (L-smoothness ⇒ Hessian upper bound)**.** *If $f \in C^2$ and f is globally L-smooth, then $\langle \nabla^2 f(x)\, u, u\rangle \le L\|u\|^2$ for all $x, u \in E$.*

Proof. Apply Theorem 31 with the fact that ∇f has Fréchet derivative $\nabla^2 f(x)$ at each x (Lemma 78). □

Lemma 83 (Operator-norm bound for symmetric PSD operators with quadratic-form bound)**.** *Let $A : E \to_L E$ be self-adjoint with $0 \le \langle Au, u\rangle$ and $\langle Au, u\rangle \le c\|u\|^2$ for all u, where $0 \le c$. Then $\|A\| \le c$.*

Proof. By the polarization identity (using self-adjointness),

$$4\,\langle Au, v\rangle = \langle A(u+v), u+v\rangle - \langle A(u-v), u-v\rangle.$$

Combining with $0 \le \langle Aw, w\rangle \le c\|w\|^2$ gives $|\langle Au, v\rangle| \le \frac{c}{2}\,(\|u\|^2 + \|v\|^2)$. Substituting $v = (\|u\|/\|Au\|)\, Au$ yields $\|Au\| \le c\|u\|$, hence $\|A\| \le c$. □

Lemma 84 (Self-adjointness of the Hessian)**.** *If $f \in C^2$, then for every $x, u, v \in E$:*

$$\langle \nabla^2 f(x)\, u,\, v\rangle = \langle u,\, \nabla^2 f(x)\, v\rangle.$$

Proof. By the symmetry of the second Fréchet derivative (`ContDiffAt.isSymmSndFDerivAt`), $(D^2 f(x))(u)(v) = (D^2 f(x))(v)(u)$. Combining with the Riesz duality used to define the gradient (∇f is the composition of Df with $(\text{InnerProductSpace.toDual})^{-1}$) and the chain rule transfers the symmetry to the Hessian. □

Lemma 85 (Hessian upper bound + convexity ⇒ L-smoothness)**.** *Suppose $f \in C^2$ is convex on E and $\langle \nabla^2 f(x)\, u, u\rangle \le L\|u\|^2$ for all x, u. Then f is globally L-smooth.*

Proof. By Theorem 30, it suffices to bound $\|\nabla^2 f(x)\| \le L$ for all x. The Hessian is self-adjoint by Lemma 84. Since f is convex, it is strongly convex with modulus 0, so Lemma 79 gives $\langle \nabla^2 f(x)\, u, u\rangle \ge 0$. Conclude by Lemma 83. □

Corollary 86 (Hessian sandwich characterization of L-smooth convex functions)**.** *Suppose $f \in C^2$ is convex on E. Then f is globally L-smooth if and only if $\langle \nabla^2 f(x)\, u, u\rangle \le L\|u\|^2$ for all $x, u \in E$, i.e. $\nabla^2 f(x) \preceq LI$.*

Proof. Direct from Lemmas 82 and 85. □

Lemma 87 (Inner-product expansion of $\|a + r\,b\|^2$)**.** *For $a, b \in E$ and $r \in \mathbb{R}$:*

$$\|a + r\,b\|^2 = \|a\|^2 + 2r\,\langle a, b\rangle + r^2\,\|b\|^2.$$

Theorem 88 (Strong convexity ⇒ global PL)**.** *If f is strongly convex on E with modulus $m > 0$ and Fréchet differentiable on E, then f is globally m-PL:*

$$\|\nabla f(x)\|^2 \;\ge\; 2m\,(f(x) - \inf f) \quad \textit{for all } x \in E.$$

Proof. Fix x and set $g := \nabla f(x)$. By the first-order lower bound (Lemma 75), for every $z \in E$,

$$f(z) \;\geq\; f(x) + \langle g,\, z - x\rangle + \tfrac{m}{2}\,\|z - x\|^2.$$

Expanding $0 \leq \|g + m(z - x)\|^2$ via Lemma 87 gives

$$0 \;\leq\; \|g\|^2 + 2m\,\langle g,\, z - x\rangle + m^2\,\|z - x\|^2.$$

Multiplying the first inequality by $2m > 0$ and adding to the second yields

$$f(x) - \tfrac{1}{2m}\,\|g\|^2 \;\leq\; f(z) \quad \text{for all } z \in E.$$

Hence f is bounded below and $\inf f \geq f(x) - \frac{1}{2m}\|g\|^2$, which rearranges to $\|g\|^2 \geq 2m\,(f(x) - \inf f)$. ∎